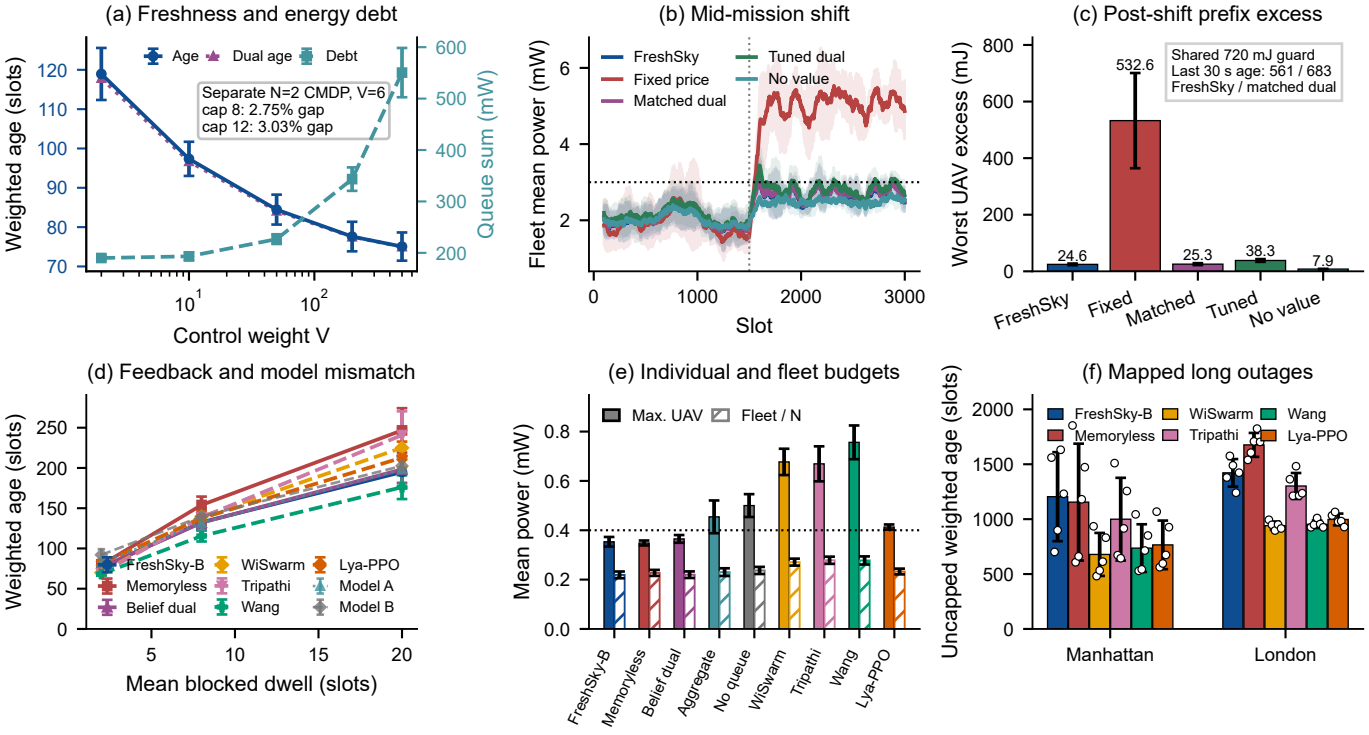

(a) Freshness and energy debt
Age
Dual age
Debt
Separate N=2 CMDP, V=6
cap 8: 2.75% gap
cap 12: 3.03% gap
Weighted age (slots)
Queue sum (mW)
Control weight V
(b) Mid-mission shift
FreshSky
Fixed price
Matched dual
Tuned dual
No value
Fleet mean power (mW)
Slot
(c) Post-shift prefix excess
Shared 720 mJ guard
Last 30 s age: 561 / 683
FreshSky / matched dual
Worst UAV excess (mJ)
532.6
24.6
25.3
38.3
7.9
FreshSky
Fixed
Matched
Tuned
No value
(d) Feedback and model mismatch
Weighted age (slots)
FreshSky-B
Memoryless
Belief dual
WiSwarm
Tripathi
Wang
Lya-PPO
Model A
Model B
Mean blocked dwell (slots)
(e) Individual and fleet budgets
Max. UAV
Fleet / N
Mean power (mW)
FreshSky-B
Memoryless
Belief dual
Aggregate
No queue
WiSwarm
Tripathi
Wang
Lya-PPO
(f) Mapped long outages
FreshSky-B
Memoryless
WiSwarm
Tripathi
Wang
Lya-PPO
Uncapped weighted age (slots)
Manhattan
London